OZ Lab

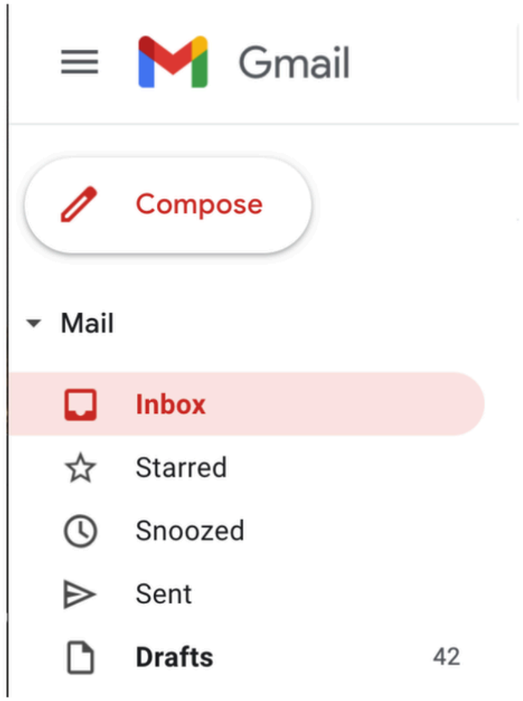
est. August 2024

#2 / Machine Learning Foundations

Welcome to Machine Learning

@oz.lab__

Learning to Recognize Spam





OZ AI Study #2

Machine Learning Applications

OZ AI Study #2

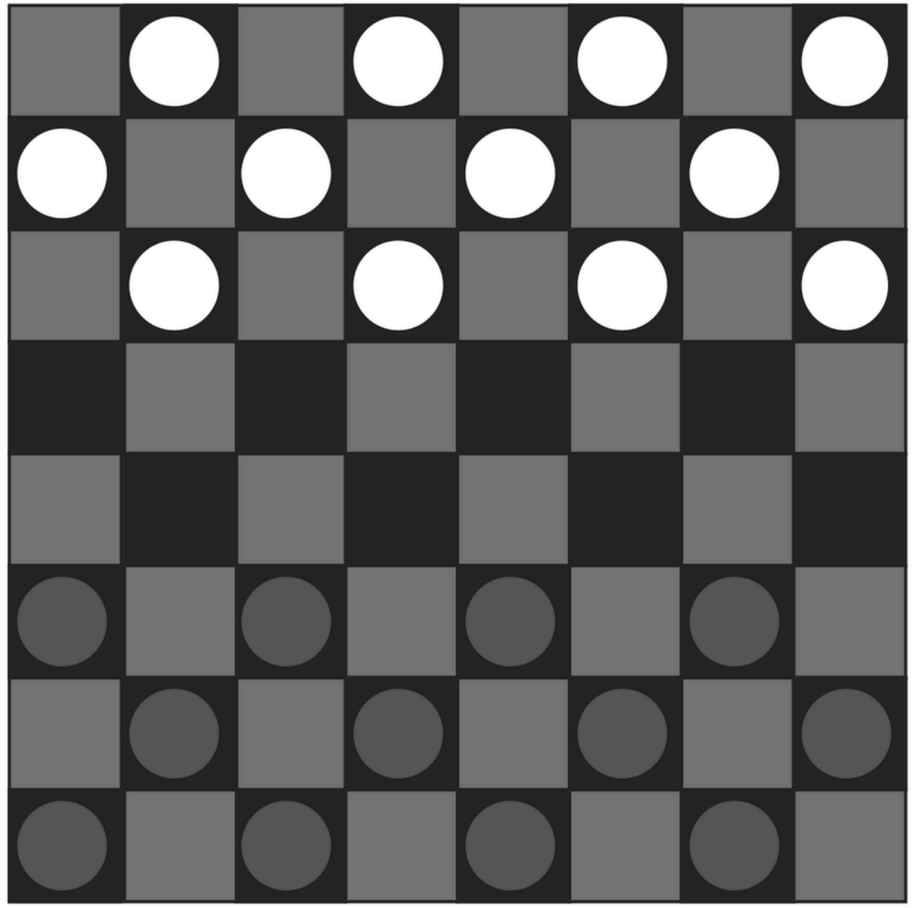
What Is Machine Learning?

Learning from Experience

Machine learning

"Field of study that gives computers the ability to learn without being explicitly programmed."

Arthur Samuel (1959)



A Map of Machine Learning

rapid advancements

used most in real-world applications

- Supervised learning ← course 1, 2

- Unsupervised learning ←
- Recommender systems
- Reinforcement learning

course 3



Practical advice for applying learning algorithms ↑

OZ AI Study #2

Supervised Learning

Learning from Inputs and Labels







input



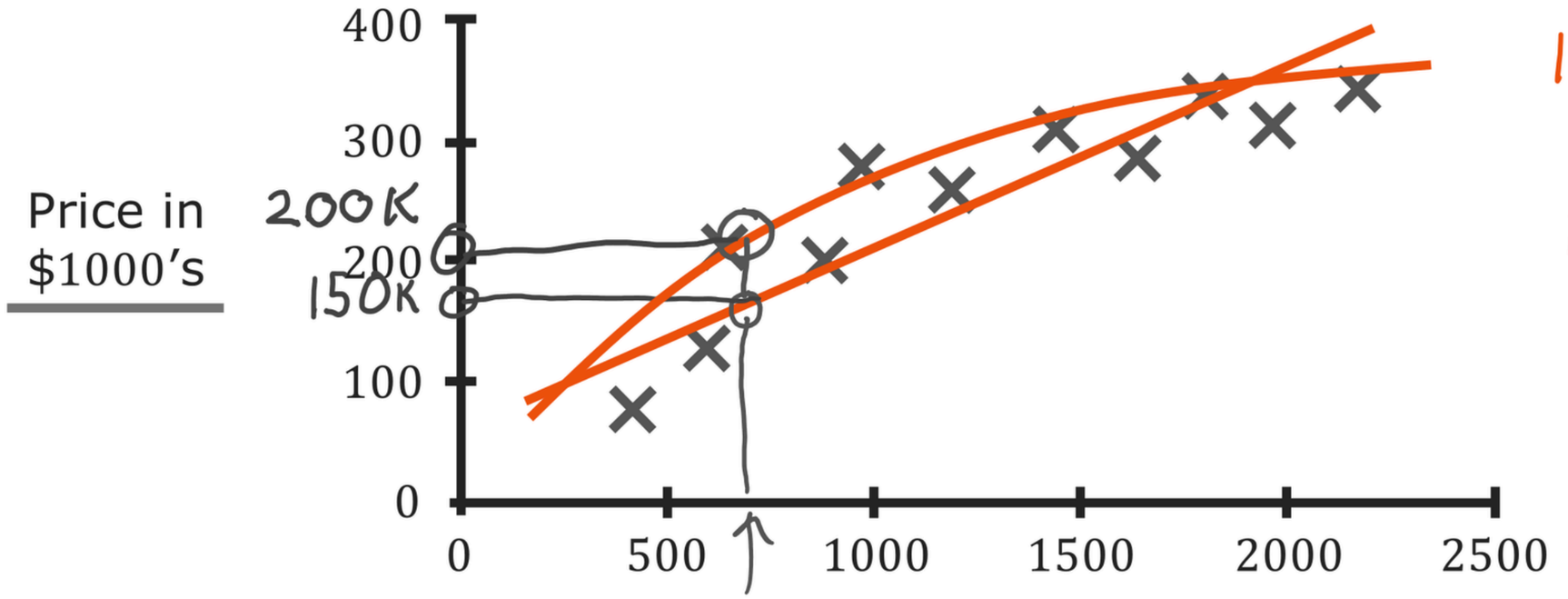
output label

Learns from being given “right answers”

Examples of Supervised Learning

Input (X)		Output (Y)	Application
email	→	spam? (0/1)	spam filtering
audio	→	text transcripts	speech recognition
English	→	Spanish	machine translation
ad, user info	→	click? (0/1)	online advertising
image, radar info	→	position of other cars	self-driving car
image of phone	→	defect? (0/1)	visual inspection

Regression Predicts Numbers



Regression

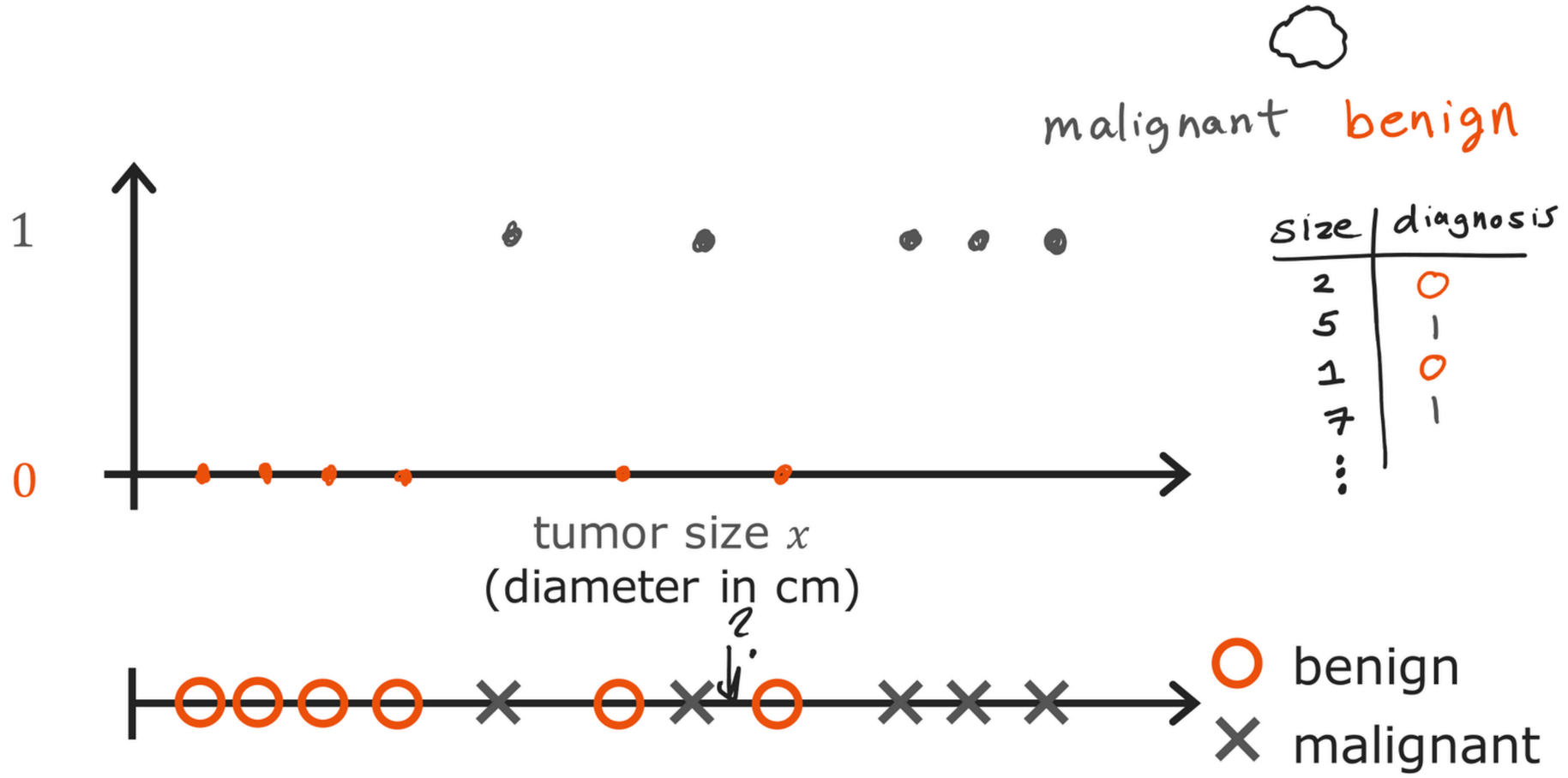
Predict a number

infinitely many possible outputs

OZ AI Study #2

Classification

Predicting One of Two Classes



Classification with More Classes

benign

malignant type1

malignant type 2



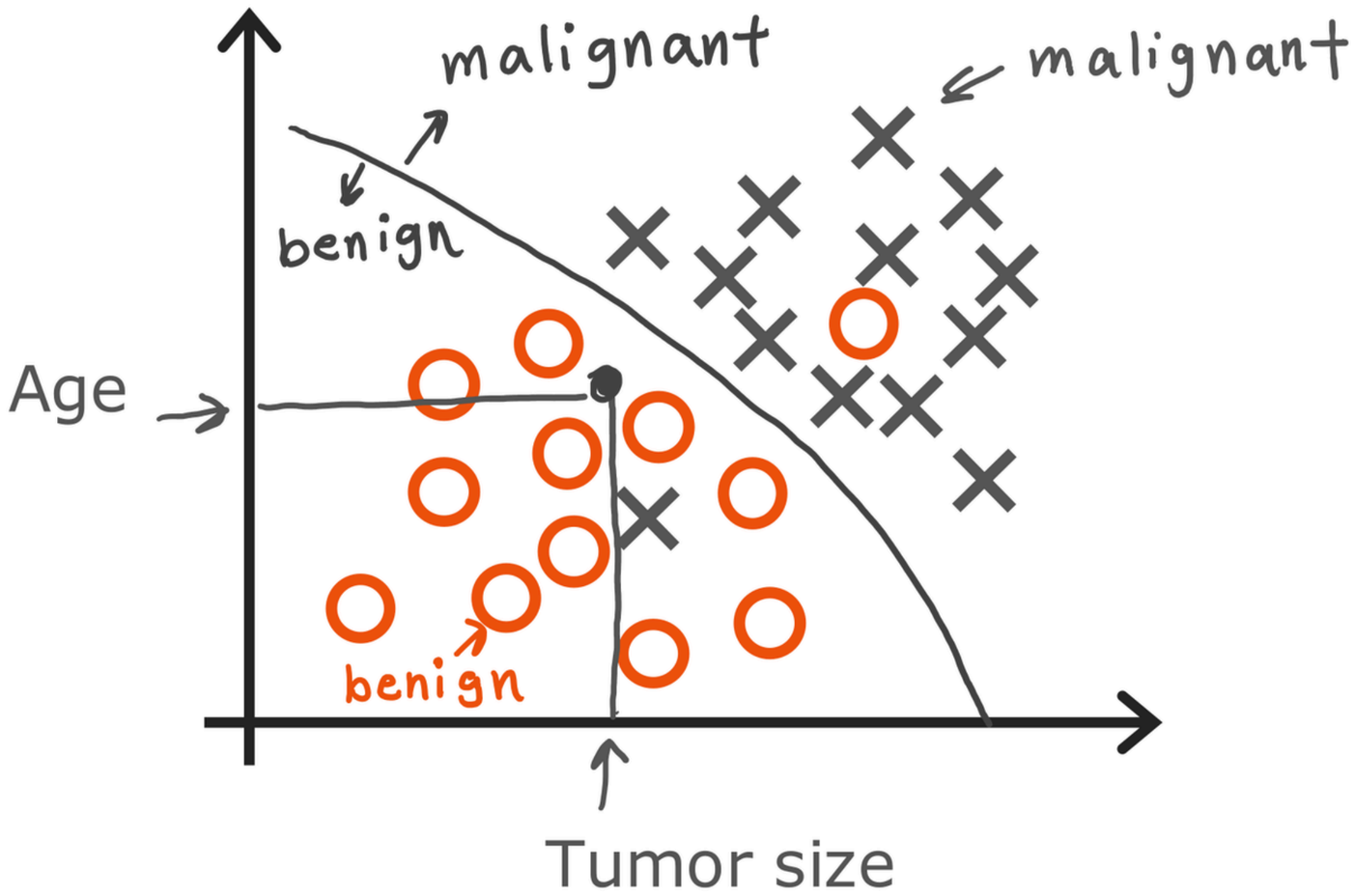
class category

Classification

predict categories cat dog benign malignant 0, 1, 2

small number of possible outputs

Using More than One Input



Regression and Classification

Learns from being given “right answers”

Regression

Predict a number

infinitely many possible outputs

Classification

predict categories

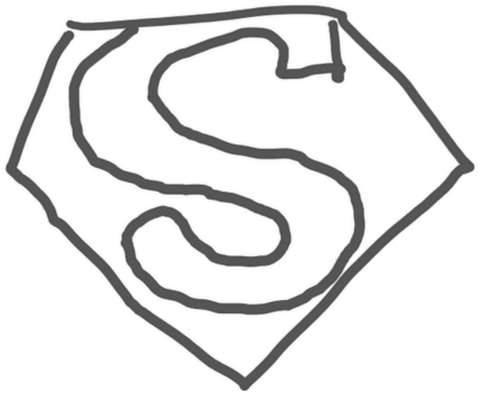
small number of possible outputs

OZ AI Study #2

Unsupervised Learning

From Labels to Unlabeled Data

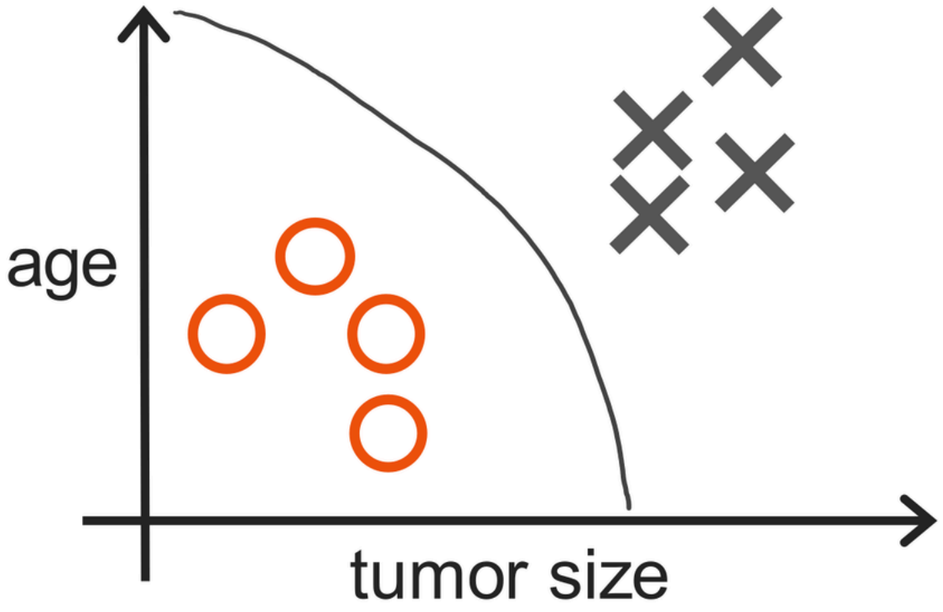
Previous: Supervised learning



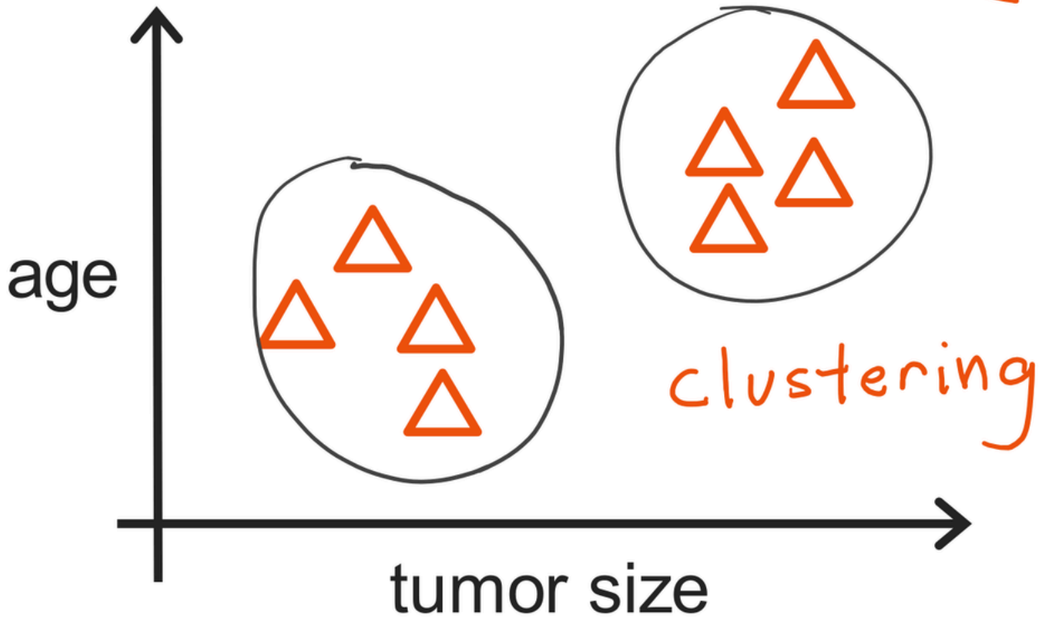
Now: Unsupervised learning

Learning with and without Labels

Supervised learning
Learn from data labeled with the "right answers"



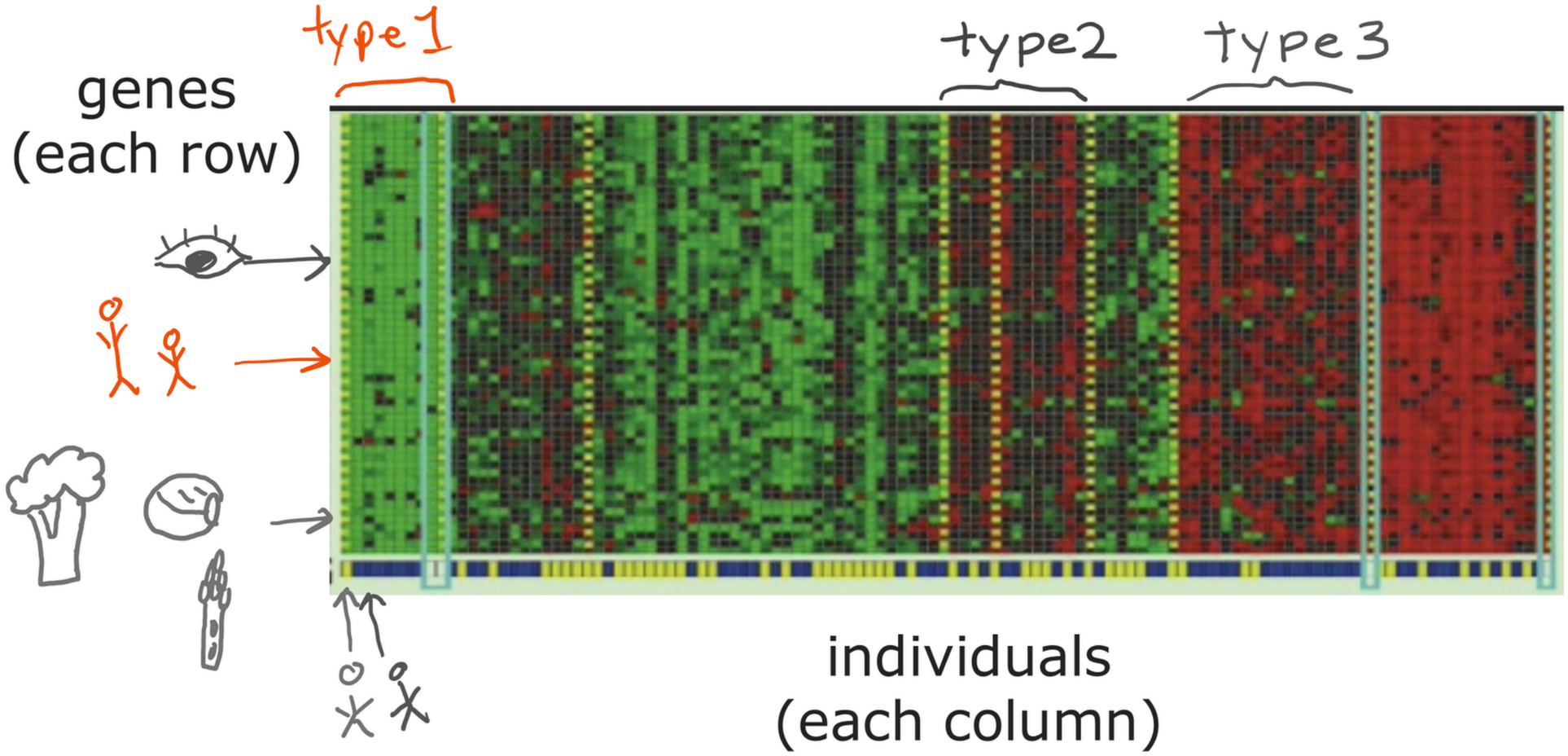
Unsupervised learning
Find something interesting in unlabeled data.



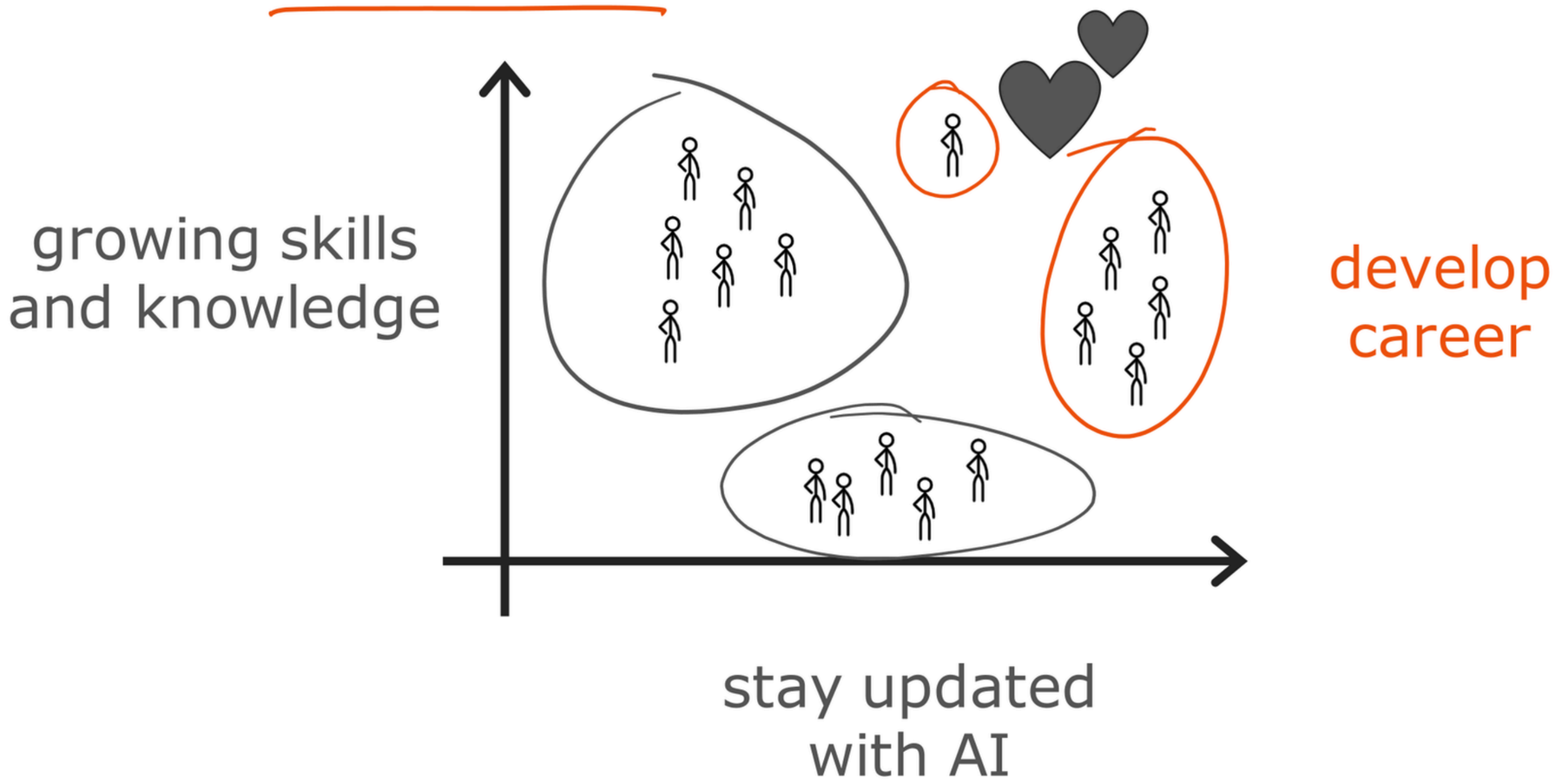
Grouping News by Story



Finding Groups in Gene Data



Grouping Customers by Interests



OZ AI Study #2

Beyond Clustering

Clustering, Anomalies, and Compression

Data only comes with inputs x, but not output labels y.
Algorithm has to find structure in the data.

Clustering
Group similar data points together.

Dimensionality reduction
Compress data using fewer numbers.

Anomaly detection
Find unusual data points.

Which Problems Need Labels?

Of the following examples, which would you address using an unsupervised learning algorithm?

- ☐ Given email labeled as spam/not spam, learn a spam filter.
- ☑ Given a set of news articles found on the web, group them into sets of articles about the same story.
- ☑ Given a database of customer data, automatically discover market segments and group customers into different market segments.
- ☐ Given a dataset of patients diagnosed as either having diabetes or not, learn to classify new patients as having diabetes or not

OZ AI Study #2

Linear Regression with One Feature

Predicting House Prices



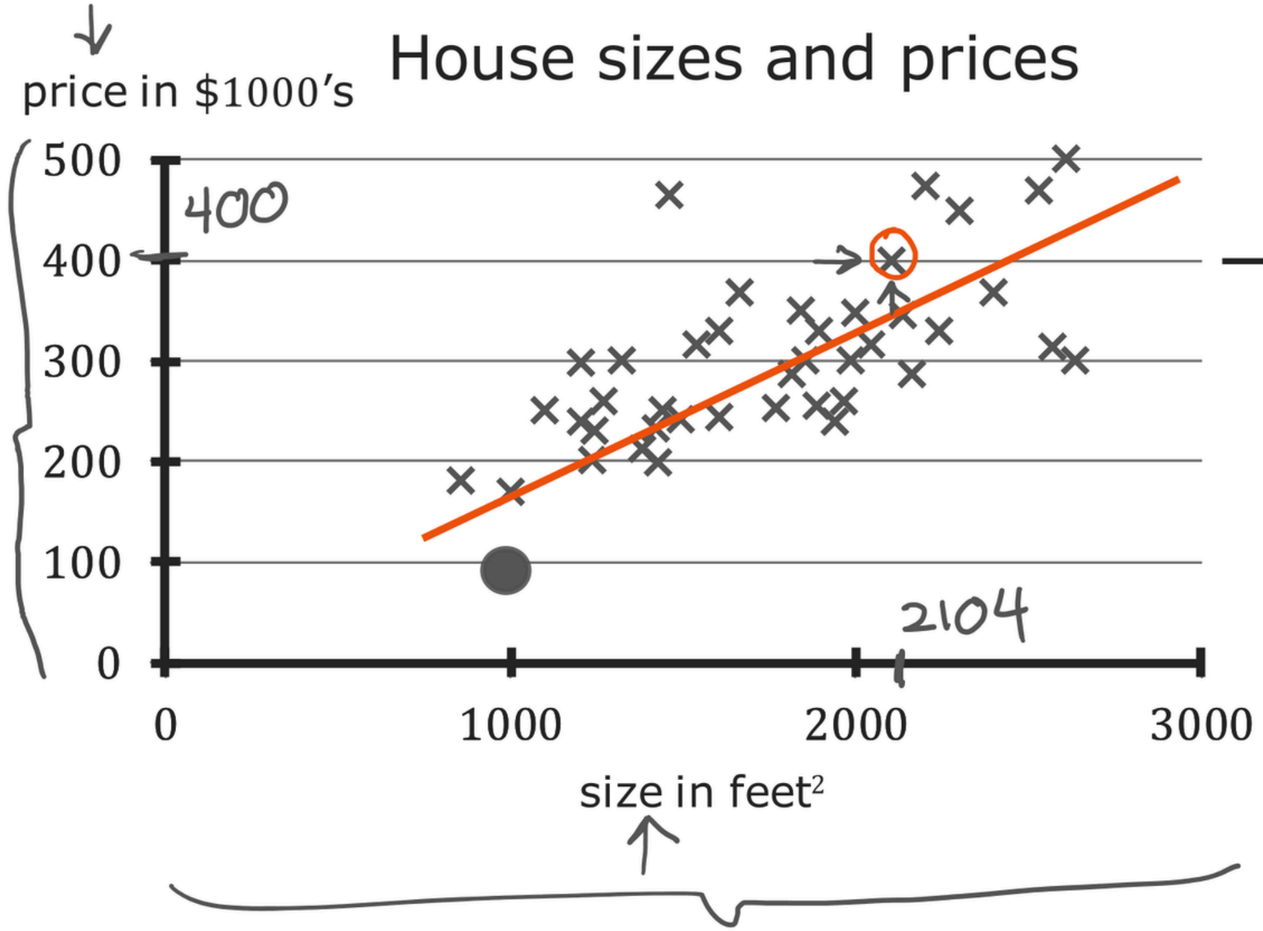
categories
cat
dog
2
disease
10

Regression model
Predicts numbers
Infinitely many possible outputs

Supervised learning model
Data has "right answers"

Classification model
Predicts categories
Small number of possible outputs

From a Plot to a Data Table



Data table

size in feet²	price in $1000's
2104	400
1416	232
1534	315
852	178
...	...
3210	870

Reading Training Data Notation

Terminology

Training set: Data used to train the model

	x size in feet^2	y price in \$1000's
(1)	2104	400
(2)	1416	232
(3)	1534	315
(4)	852	178
...	...	...
(47)	3210	870

$m = 47$

$x^{(1)} = 2104 \quad y^{(1)} = 400$

$(x^{(1)}, y^{(1)}) = (2104, 400)$

$x^{(2)} = 1416 \quad y^{(2)} \neq x^2$ not exponent

Notation:

x = "input" variable
feature

y = "output" variable
"target" variable

m = number of training examples

(x, y) = single training example

$(x^{(i)}, y^{(i)})$

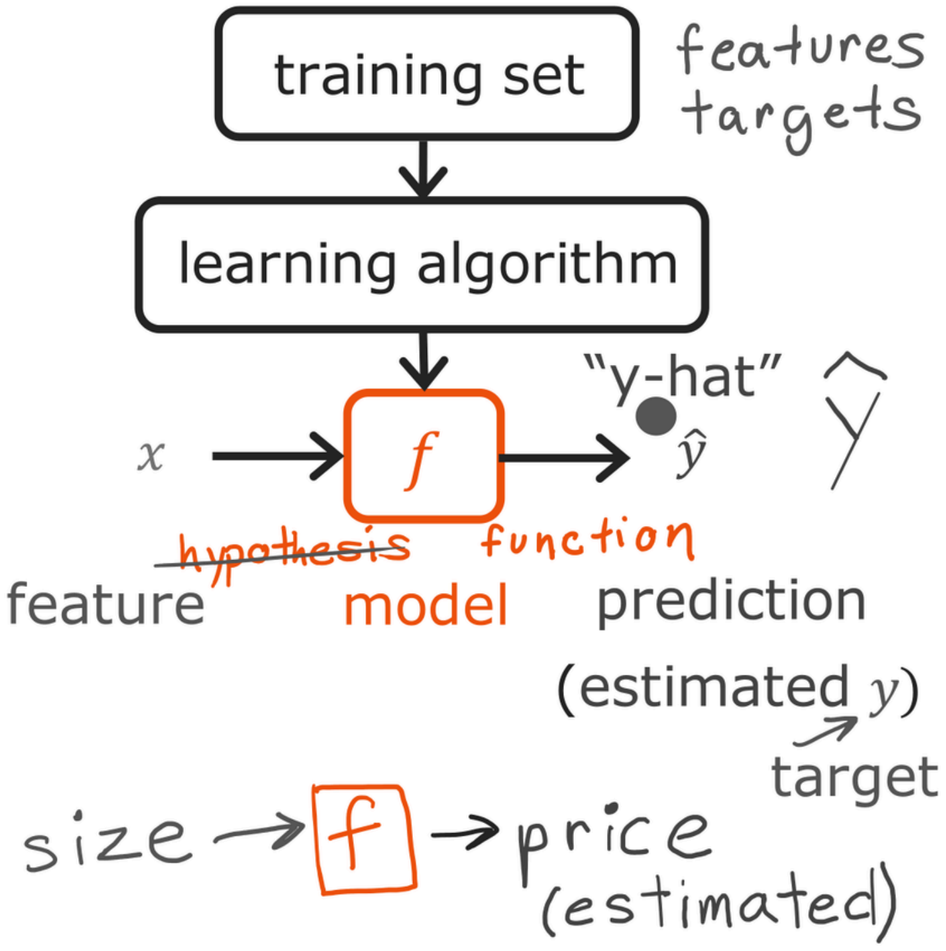
$(x^{(i)}, y^{(i)})$ = i^{th} training example

index (1st, 2nd, 3rd ...)

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Using a Linear Regression Model

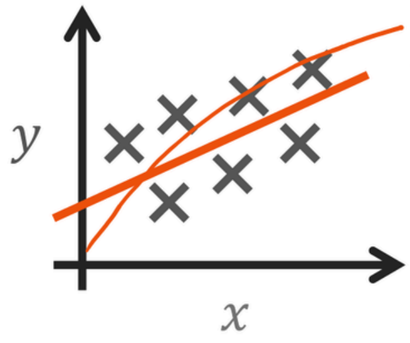
Training a Model to Make Predictions



How to represent f?

$f_{w,b}(x) = wx + b$

$f(x)$



single feature x

Linear regression with one variable.

size

Univariate linear regression.

one variable

OZ AI Study #2

The Cost Function

The Two Parameters of a Line

features size in feet^2 (x)	targets price \$1000's (y)
2104	460
1416	232
1534	315
852	178
...	...

Model: $f_{w,b}(x) = wx + b$

w, b: parameters

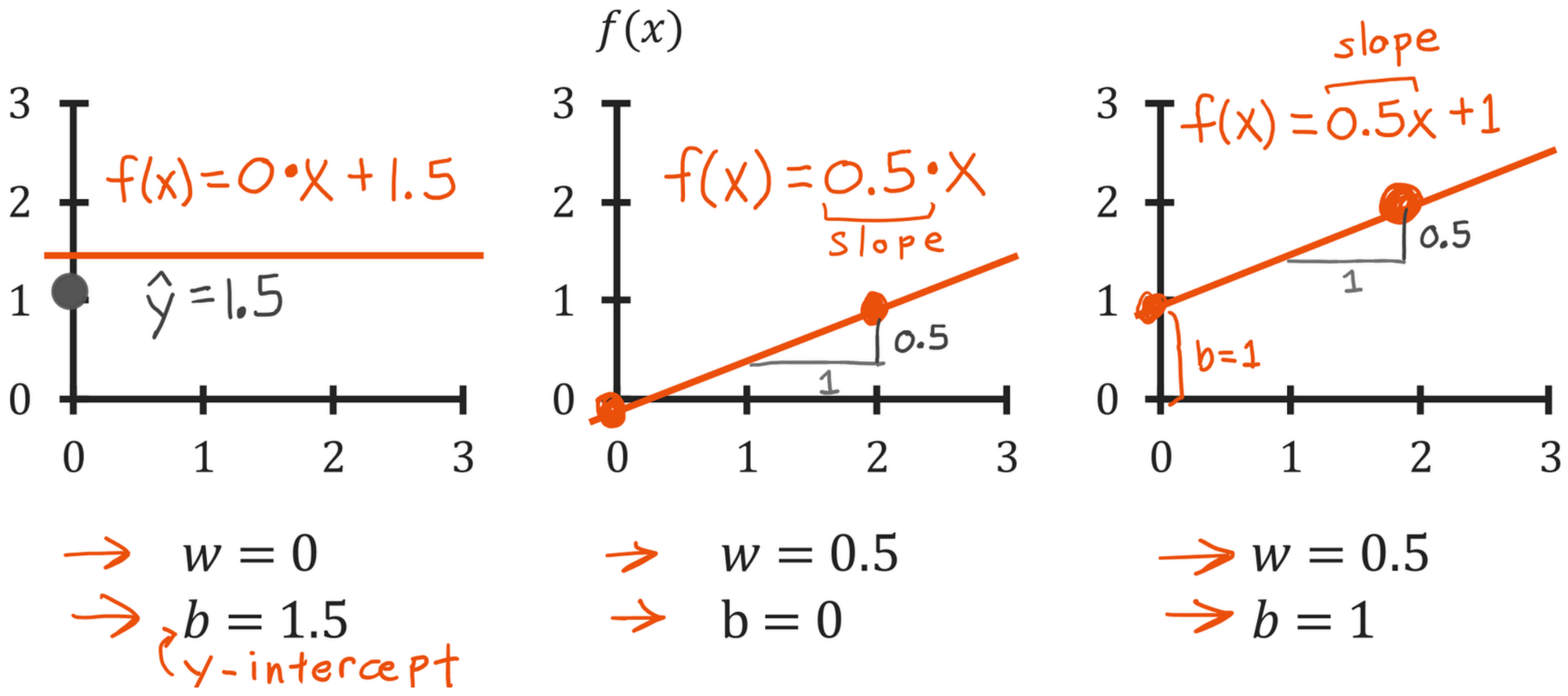
coefficients

weights

What do w, b do?

Changing the Slope and Intercept

$$f_{w,b}(x) = wx + b$$

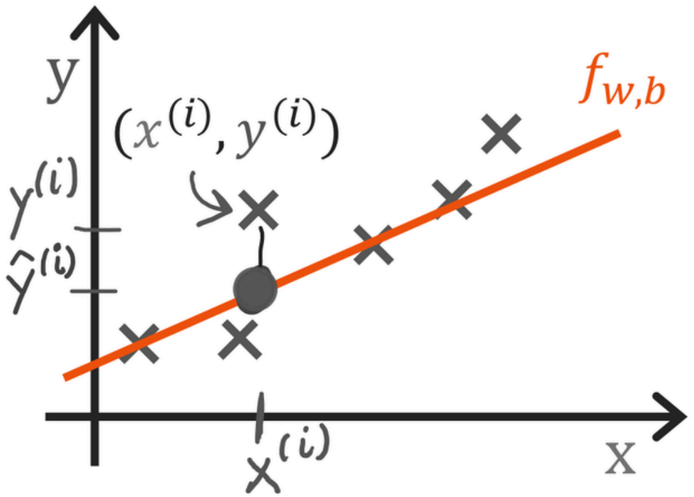


→ $w = 0$
→ $b = 1.5$ (y-intercept)

→ $w = 0.5$
→ $b = 0$

→ $w = 0.5$
→ $b = 1$

The Squared Error Cost Function



$$\hat{y}^{(i)} = f_{w,b}\left(x^{(i)}\right) \leftarrow$$

$$f_{w,b}\left(x^{(i)}\right) = wx^{(i)} + b$$

Cost function: Squared error cost function

$$J(w,b) = \frac{1}{2m}\sum_{i=1}^{m}\left(\hat{y}^{(i)} - y^{(i)}\right)^2$$

error

m = number of training examples

$$J(w,b) = \frac{1}{2m}\sum_{i=1}^{m}\left(f_{w,b}\left(x^{(i)}\right) - y^{(i)}\right)^2$$

intuition (next!)

Find w, b:

$\hat{y}^{(i)}$ is close to $y^{(i)}$ for all $(x^{(i)}, y^{(i)})$.